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PAPER

Feature fusion based on global-local weighted attention model for automatic epileptic seizure detection

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E-mail: zxhklwx@163.com, chenshixiong@cuhk.edu.cn, 224050069@link.cuhk.edu.cn, kezhang@cug.edu.cn, wangxin@siat.ac.cn, 123090850@link.cuhk.edu.cn, 224050024@link.cuhk.edu.cn and 458260329@qq.com**Keywords:** epilepsy detection, EEG analysis, global-local weighted attention (GLWA), feature fusion**Abstract**

Objective. Epilepsy is a neurological disorder characterized by recurrent seizures, which present significant challenges in both diagnosis and treatment. Despite advances in seizure detection, existing methods often struggle with accurately capturing the complex and dynamic interactions between temporal, spatial, and spectral features of electroencephalography (EEG) signals. This leads to limitations in the detection accuracy and generalization across different datasets. **Approach.** To address these challenges, we propose global-local weighted attention (GLWA) model, which integrates temporal, spatial, and spectral features through a local-global attention mechanism. At the same time, GLWA effectively balances both global and local features, capturing comprehensive information from EEG signals to enhance seizure detection accuracy. **Main results.** Our proposed model achieves accuracy rates of 98.82% and 98.89% on the CHB-MIT and Siena datasets, respectively. These results demonstrate the model's capability to effectively integrate these features, resulting in improved detection performance. **Significance.** Furthermore, we visualize the model's decision-making process to gain insights into the attention distribution across different brain regions and spectral frequency bands, further emphasizing GLWA's potential in seizure detection. This work demonstrates the model's superior performance and interpretability, providing a robust approach for accurate and generalizable identification of seizures.

1. Introduction

Epilepsy is a neurological disorder characterized by recurrent, unprovoked seizures [1]. It affects over 70 million people worldwide and is considered one of the most common and serious brain disorders [2]. The unpredictability and severity of seizures can significantly impact the quality of life for individuals with epilepsy. Accurate and temporally detection of seizures is crucial for managing epilepsy and preventing potential harm. Advances in

electroencephalography (EEG) have made it possible to monitor brain activity in real-time, providing a valuable tool for detecting seizures [3]. However, automatic seizure detection from EEG signals remains a challenging task due to the complex and dynamic nature of brain activity.

Seizure detection is further complicated by several practical challenges. First, the process of analyzing EEG data traditionally requires trained experts, who are often in short supply, leading to delays in diagnosis and treatment [4]. Second, the

manual interpretation of EEG recordings is temporal-consuming and prone to errors, particularly when analyzing long-term data [5]. These factors underscore the need for automated and efficient seizure detection systems that can alleviate the reliance on human expertise while delivering accurate and temporally results.

In recent years, epileptic seizure detection has increasingly become a focal point of research [3]. Machine learning (ML) [6] methods have emerged as promising solutions to these challenges. By automating the feature extraction and classification processes, ML models have significantly improved the efficiency and accuracy of seizure detection. Traditional ML approaches often rely on handcrafted features extracted from temporal or spectral domains, which are subsequently fed into classifiers such as support vector machines (SVMs) [7], random forests, or k-nearest neighbors (KNNs) [8, 9]. Similarly, decision trees have been applied to EEG datasets to enhance classification accuracy [10]. Although these methods mark progress, their dependence on manually crafted features restricts their capacity to generalize across different datasets or accommodate the intricate nature of EEG signals.

Many deep learning-based methods, such as convolutional neural networks (CNNs) [11–13] and long short-term memory networks (LSTMs) [14–16] have achieved good results. In addition, scholars have tried to integrate attention mechanisms with these models to further improve performance. For example, Tang *et al* [17] proposed an automatic epilepsy detection method based on bidirectional LSTM and path signature, which enhances the model's ability to capture the temporal dependencies of EEG signals through attention mechanisms, thereby improving detection accuracy. Another study [18] introduced a personalized approach using CNNs and gated recurrent (GRU) neural networks, which effectively selects EEG channels through attention mechanisms, reducing the number of required electrodes and making it easier for patients to wear them for extended periods. At the same time, many studies have attempted to combine multidimensional features such as temporal, spatial, and spectral to improve the effectiveness of seizure prediction. Wang *et al* [19] proposed a spatiotemporal channel attention residual network, which significantly improves the performance of epilepsy detection by combining temporal and spatial features. In addition, GlepNet [20] is better able to capture local and global epileptic features in EEG signals by combining temporal convolution and multi-head attention mechanisms. Although the existing research has made remarkable progress, how to integrate spatio-temporal-spectral features more comprehensively to improve the robustness and generalization ability of detection is still an urgent problem to be solved.

To overcome these limitations, the integration of features across temporal, spatial, and spectral domains is crucial [21]. A comprehensive fusion of these domains allows models to better reflect the underlying seizure dynamics and improve detection accuracy and robustness [22]. Beyond this, a weighted fusion mechanism that dynamically balances the contributions of global and local attention features further enhances model performance [23]. Such a mechanism enables the effective capture of localized variations, such as brief oscillatory patterns critical for seizure detection, while maintaining a global perspective to monitor seizure activity across the entire EEG signal. This approach not only addresses the shortcomings of single-modality reliance, such as the inability to fully capture seizure dynamics from one domain alone, but also ensures that global and local features are seamlessly integrated. This allows the model to leverage both localized information indicative of seizures and broader global patterns for dynamic monitoring. However, there remains a need to refine these mechanisms further and validate their generalization across diverse datasets.

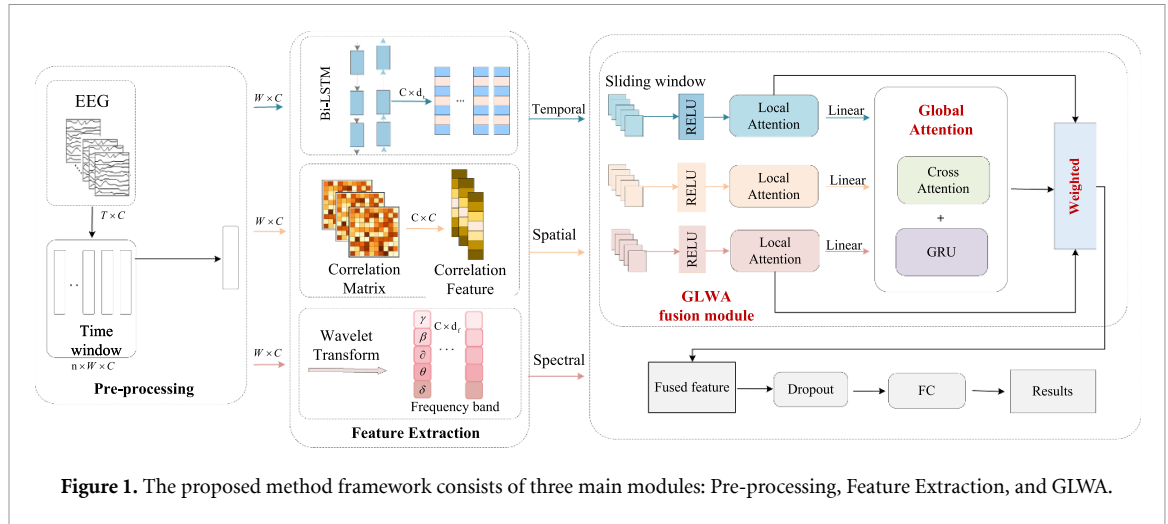
To address these challenges, this work introduces a novel seizure detection framework named global-local weighted attention (GLWA) that resolves key limitations in existing methods. Specifically, it addresses the problem of missing critical seizure patterns due to single-modality reliance and the inadequate integration of global and local features. The framework is built on the following key contributions:

- We extract features from three dimensions: temporal, spatial, and spectral, to achieve a comprehensive fusion of information, enabling the model to effectively detect seizures.
- We propose a new model for feature fusion, GLWA, which integrates different features through local-global attention within windows, balancing window-specific global and local features via weighted fusion.
- Extensive evaluations on the CHB-MIT and Siena datasets, demonstrating state-of-the-art sensitivity, specificity, and overall classification accuracy, while validating the robustness and generalization of the proposed framework across diverse patient populations.

The remainder of the paper is organized as follows: section 2 presents the methods, section 3 covers experiments and datasets, section 4 discusses the results, and section 5 concludes with future work.

2. Methods

The complete model framework, as illustrated in figure 1, consists of three main components:



Algorithm 1. Overall Model Framework.

- 1: **Input:** EEG signal X , segmented into sliding windows of size W
 - 2: **Output:** Fused feature representation z_{fusion}
 - 3: **Step 1: Feature Extraction for Each Window W_i**
 - 4: **for** $W_i \in X$ **do**
 - 5: Temporal features: $z_{\text{temporal}} \leftarrow \text{Bi-LSTM}(W_i)$
 - 6: Spatial features: $z_{\text{spatial}} \leftarrow \text{Attention}(\text{Correlation}(W_i))$
 - 7: Spectral features: $z_{\text{spectral}} \leftarrow \text{Attention}(\text{Wavelet}(W_i))$
 - 8: **end for**
 - 9: **Step 2: Local Attention for Each Feature**
 - 10: **for** $z \in \{z_{\text{temporal}}, z_{\text{spatial}}, z_{\text{spectral}}\}$ **do**
 - 11: **for** $W_i \in z$ **do**
 - 12: Local attention weights: $A_{\text{local}}^{W_i} \leftarrow \text{Softmax}\left(\frac{Q^{W_i}(K^{W_i})^T}{\sqrt{d}}\right)$
 - 13: Local attention output: $z_{\text{local}}^{W_i} \leftarrow A_{\text{local}}^{W_i} V^{W_i}$
 - 14: **end for**
 - 15: Combine outputs: $z_{\text{local}} \leftarrow \text{Concat}(\{z_{\text{local}}^{W_i}\}_{i=1}^N)$
 - 16: **end for**
 - 17: **Step 3: Global Attention Across Modalities**
 - 18: Global attention inputs: $Q \leftarrow W_q z_{\text{temporal}}, K \leftarrow W_k z_{\text{spatial}}, V \leftarrow W_v z_{\text{spectral}}$
 - 19: Global attention weights: $A_{\text{global}} \leftarrow \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)$
 - 20: Global attention output: $z_{\text{global}} \leftarrow A_{\text{global}} V$
 - 21: **Final Output:** $z_{\text{fusion}} \leftarrow z_{\text{global}}$
-

preprocessing, feature extraction, and the GLWA module. Firstly, during the preprocessing stage, EEG signals are filtered using a 0.5–50 Hz bandpass filter to remove noise and artifacts. Secondly, feature extraction is carried out across the temporal, spatial, and spectral domains to capture diverse signal features. Finally, the GLWA module combines the extracted features through a two-stage attention mechanism: local attention, which operates within sliding windows to capture fine-grained details, and global attention, which integrates contextual information across the entire signal. The outputs are then fused using a weighted mechanism and classified using a softmax layer. The following sections provide a detailed explanation of each component, based on the GLWA method's algorithmic flowchart in algorithm 1.

2.1. Feature extraction

The EEG signals are first preprocessed and segmented into time windows, with each window covering 1 second of signal data. Assuming a sampling rate of f_s , each window contains f_s temporal points. Feature extraction is then performed on a per-window basis for temporal, spatial, and spectral domains as described below:

2.1.1. Temporal feature extraction

Temporal features are extracted using a bidirectional LSTM (Bi-LSTM) network to model temporal dependencies within each sliding window. For a window $W \in \mathbb{R}^{C \times T}$, where $T = f_s$ is the number of temporal steps and C is the number of channels, the forward and backward hidden states are

computed as follows:

$$\text{fh}_t = \text{LSTM}_{\text{forward}}(x_t, \text{fh}_{t-1}), \quad (1)$$

$$\text{bh}_t = \text{LSTM}_{\text{backward}}(x_t, \text{bh}_{t+1}), \quad (2)$$

where $t \in [1, T]$. The final temporal feature representation is obtained by concatenating the forward and backward hidden states:

$$z_{\text{temporal}}^W = [\text{fh}_t; \text{bh}_t] \in \mathbb{R}^{C \times d_t}. \quad (3)$$

By the way, Bi-LSTM effectively captures bidirectional temporal dependencies in EEG signals, crucial for modeling dynamic patterns in epilepsy detection. Its ability to process both past and future contexts within each time window enables robust representation of non-stationary EEG dynamics, such as transient spikes or slow waves, which are feature of epileptic seizures [24].

2.1.2. Spatial feature extraction

Spatial features are extracted to capture correlations between EEG channels. For each sliding window W , a Pearson correlation matrix [25] $P \in \mathbb{R}^{C \times C}$ is constructed:

$$P_{uv} = \frac{\text{Cov}(W_u, W_v)}{\sigma_{W_u} \sigma_{W_v}}, \quad (4)$$

where W_u, W_v are the signals of the u th and v th channels within the window, Cov denotes covariance, and σ represents standard deviation.

The correlation matrix is then flattened into a vector:

$$P_{\text{flat}} \in \mathbb{R}^{C^2}. \quad (5)$$

This flattened vector is passed through a fully connected layer to extract spatial features:

$$z_{\text{spatial}}^W = W_f P_{\text{flat}} + b_f \in \mathbb{R}^{C \times d}, \quad (6)$$

where W_f is the weight matrix of the fully connected layer, b_f is the bias term, and d is the output feature dimension. The Pearson correlation matrix effectively captures inter-channel spatial dependencies in EEG signals, essential for modeling synchronized brain activity patterns during epileptic seizures [26].

2.1.3. Spectral feature extraction

Spectral features are extracted using wavelet transform [27] to decompose the EEG signals into multiple spectral bands. For each sliding window W , the wavelet transform of the u th channel is given by:

$$W_u^f = \int_{-\infty}^{\infty} W_u(t) \psi_f^*(t) dt, \quad f \in \{f_1, f_2, \dots, f_n\}, \quad (7)$$

where f represents the spectral band, and ψ_f^* is the wavelet basis function. We initialize the frequency bands to 5, corresponding to $\delta, \theta, \alpha, \beta$, and γ .

The final spectral representation is obtained by directly summing over the spectral bands without attention:

$$z_{\text{spectral}}^W = \sum_f F_f \in \mathbb{R}^{C \times d_f}, \quad (8)$$

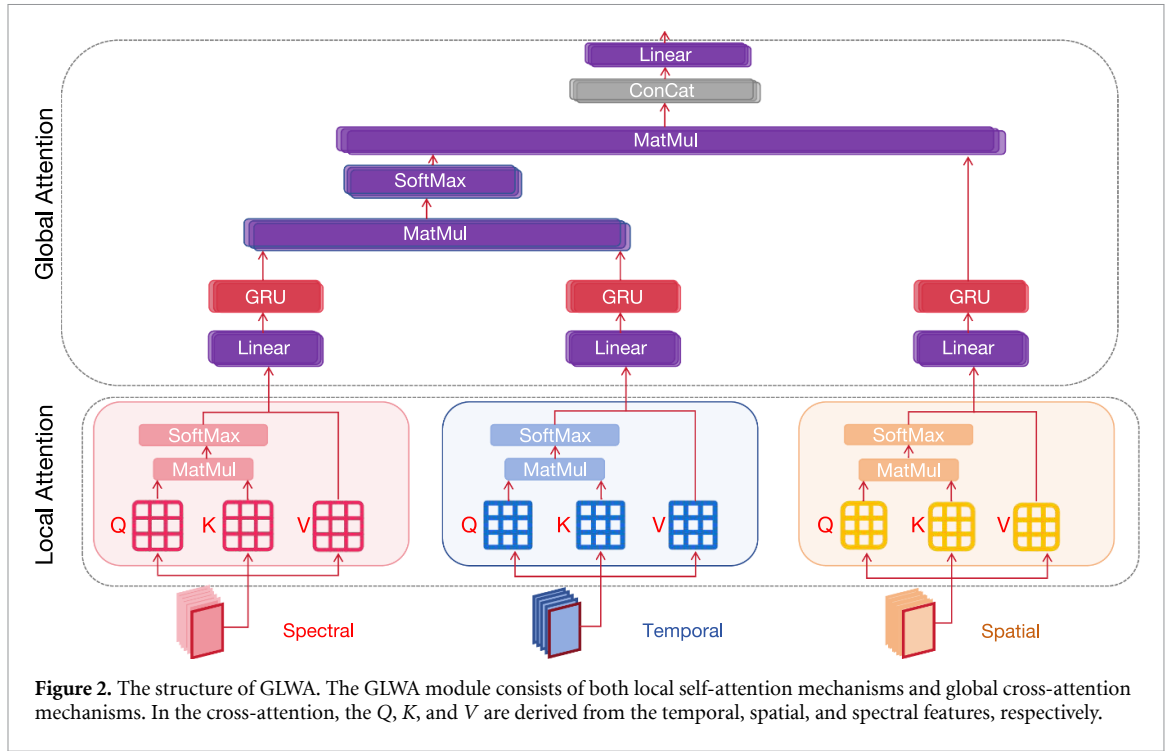
where F_f represents the features of the f -th spectral band, and the sum is taken across all spectral bands. The wavelet transform is particularly effective for EEG analysis because it uniquely captures both time and frequency information simultaneously [28]. Unlike traditional methods, it can precisely track sudden changes in brain activity while maintaining clear frequency resolution. This special capability makes it ideal for detecting the brief but distinct spectral patterns that occur during epileptic seizures [29].

2.2. GLWA

The GLWA module combines both local and global attention mechanisms to capture fine-grained details within sliding windows and long-range dependencies across modalities. This module processes three main modalities: temporal, spatial, and spectral, using a sliding window approach for attention computation. Specifically, the local attention mechanism is designed to capture fine-grained dependencies within each sliding window W , where the temporal, spatial, and spectral features are processed synchronously. This mechanism utilizes self-attention to compute local attention weights within each sliding window and applies weighted summation, enabling the model to focus on important patterns within each window. The features of each modality are processed through this local attention mechanism to effectively capture the local dependencies.

2.2.1. Local attention with sliding window mechanism

To effectively model localized patterns in EEG signals, we employ a local attention mechanism that operates within constrained spatial-temporal-spectral windows to capture detailed features. The local attention mechanism extracts features by focusing on specific local regions, typically defined by a sliding window, thereby emphasizing local feature correlations and reducing computational complexity. Specifically, to capture local dependencies in EEG signals, the mechanism uses a 3×3 sliding window, encompassing 9 consecutive feature vectors from the temporal, spatial, and spectral modalities, with a stride of 1 to generate local subsequences. In these subsequences, the local input dimensions for the temporal, spatial, and spectral modalities are all set to $C \times 64$, as the hidden dimension is established as 64 for each feature during feature processing, where C represents the number of channels. The input features of each sliding window w are processed through the



self-attention mechanism to derive query (Q), key (K), and value (V) matrices. These matrices are calculated as $Q_{\text{local}} = W_q z^w$, $K_{\text{local}} = W_k z^w$, and $V_{\text{local}} = W_v z^w$, where $W_q, W_k, W_v \in \mathbb{R}^{d \times d_h}$ are learnable projection matrices.

The local attention weights are computed by taking the dot product of the query (Q) and key (K) matrices, followed by a Softmax operation. These weights are applied to the value (V) matrix to obtain the final local feature representation as a weighted sum of the values.

After performing local attention for each sliding window, the local features from all windows are aggregated to form the final feature representation. This aggregation is achieved by concatenating the local features from each window, resulting in a complete feature representation for the entire sequence.

2.2.2. Global attention (cross-modal dependencies)

As shown in figure 2, global attention is designed to integrate information across modalities and sliding windows to capture long-range dependencies. It operates on the outputs of local attention and facilitates cross-modal interactions.

The Q, K, and V matrices for global attention are derived separately from temporal, spatial, and spectral modalities, where GRU is applied to each modality to capture temporal dependencies. The use of GRU allows for better capture of the temporal information in the global module, improving the model's ability to process sequential data:

$$Q_{\text{global}} = \text{GRU}(z_{\text{temporal}}) W_q, \quad (9)$$

$$K_{\text{global}} = \text{GRU}(z_{\text{spatial}}) W_k, \quad (10)$$

$$V_{\text{global}} = \text{GRU}(z_{\text{spectral}}) W_v, \quad (11)$$

where $W_q, W_k, W_v \in \mathbb{R}^{d \times d}$ are learnable projection matrices, and $z_{\text{temporal}}, z_{\text{spatial}}, z_{\text{spectral}}$ are the outputs of local attention from temporal, spatial, and spectral features, respectively, after being processed by the GRU layers to capture the temporal dependencies.

The global attention weights are calculated using a scaled dot-product attention mechanism:

$$A_{\text{global}} = \text{Softmax}\left(\frac{Q_{\text{global}} K_{\text{global}}^T}{\sqrt{d}}\right). \quad (12)$$

The global attention output aggregates features across modalities and sliding windows:

$$z_{\text{global}} = A_{\text{global}} V_{\text{global}}. \quad (13)$$

2.2.3. Weighted fusion

The final representation z_{fusion} is obtained by combining the outputs of the local and global attention mechanisms:

$$z_{\text{fusion}} = \alpha z_{\text{global}} + (1 - \alpha) \cdot \text{Concat}(z_{\text{temporal}}, z_{\text{spatial}}, z_{\text{spectral}}). \quad (14)$$

where α is a learnable parameter balancing the contributions of global and local features. We initialize α to 0.5 to provide a balanced starting point for the model training. The final output dimension of the global attention after Concatenation is $C \times 64$, where the final linear layer projects the concatenated features into a 64-dimensional space.

2.2.4. Evaluation metrics

In the task of epilepsy seizure detection, seizure segments are considered as the positive class, while non-seizure segments belong to the negative class. The following evaluation metrics are used to measure the performance of the model:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (15)$$

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (16)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (17)$$

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}} \quad (18)$$

$$\text{FPR/h} = \frac{\text{Number of False Positives}}{\text{Total Hours}} \quad (19)$$

where True Positive (TP) refers to seizure segments correctly classified as seizures, True Negative (TN) refers to non-seizure segments correctly classified as non-seizures, False Positive (FP) represents non-seizure segments misclassified as seizures, and False Negative (FN) indicates seizure segments that were missed by the model.

3. Experiments

The experiment was conducted in a Python 3.12 environment using PyTorch 2.5.1, with an RTX 4090 24GB GPU.

3.1. Datasets

3.1.1. CHB-MIT dataset

The CHB-MIT dataset [30, 31] is a widely used EEG dataset for seizure detection, collected from pediatric patients at the Children's Hospital Boston. It contains EEG recordings from 24 patients, with data from both seizure and non-seizure periods, and annotated seizure events. The data is segmented into 1 s time windows to capture the dynamic changes within each window, with a 0.5 s overlap [32] between seizure windows to increase the number of seizure samples. Non-seizure samples are collected by randomly selecting slices from files that do not contain seizure segments, which helps balance the dataset between seizure and non-seizure samples, ensuring the robustness and generalization of the model. The signals are sampled at 256 Hz. This dataset includes data from 22 subjects (5 males, aged 3–22 years; 17 females, aged 1.5–19 years), with EEG electrode placements based on the international 10–20 system. The CHB-MIT dataset extracted in this study is presented in table 1.

Table 1. Detailed information of the CHB-MIT dataset used in this work.

Case	Gender	Age	Number of seizures	Selected data	
				duration (min)	
				Seizure	Non-seizure
chb01	F	11	7	7.37	21.88
chb02	M	11	3	2.87	7.60
chb03	F	14	7	6.70	14.28
chb04	M	22	4	6.30	13.98
chb05	F	7	5	9.30	25.90
chb06	F	1.5	10	2.55	7.27
chb07	F	14.5	3	5.42	14.95
chb08	M	3.5	5	15.32	39.07
chb09	F	10	4	4.60	9.97
chb10	M	3	7	7.45	20.02
chb11	F	12	3	13.43	30.90
chb12	F	2	40	24.58	67.43
chb13	F	3	12	8.92	20.73
chb14	F	9	8	2.82	8.03
chb15	M	16	20	33.20	99.47
chb16	F	7	10	1.40	3.85
chb17	F	12	3	4.88	11.57
chb18	F	18	6	5.28	14.73
chb19	F	19	3	3.93	10.73
chb20	F	6	8	4.90	9.90
chb21	F	13	4	3.32	8.02
chb22	F	9	3	3.40	7.15
chb23	F	6	7	7.07	18.93
chb24	—	—	16	8.52	18.55
Total	—	—	198	193.53	504.91

3.1.2. Siena dataset

The Siena dataset [33] is a publicly available EEG dataset collected from adult patients with epilepsy, used extensively in seizure detection research. The dataset includes continuous EEG recordings from 14 patients (9 males and 5 females), with annotations for both seizure and non-seizure periods. Similar to the CHB-MIT dataset, all available EEG channels are used to ensure a comprehensive capture of changes in the EEG signal, thus improving the model's ability to detect seizures. The signals are sampled at 512 Hz. Most recordings include 29 channels, with the exception of participant PN10, which has 20 channels. The data is segmented into 1 s time windows, and the data within each window is used for feature extraction and analysis, enhancing the model's fine-grained detection ability for seizures, with a 0.5 s overlap between seizure windows to increase the number of seizure samples. This dataset includes both interictal and ictal EEG data, and the onset and end temporals of seizures are carefully annotated, making it well-suited for training and evaluating seizure detection models. The Siena dataset extracted in this study is presented in table 2.

3.2. Preprocessing

In the preprocessing phase, the raw EEG data underwent several steps to enhance signal quality and

Table 2. Detailed information of the Siena dataset used in this work.

Case	Gender	Age	Number of Seizures	Selected data	
				duration (s)	
				Seizure	Non-seizure
PN00	Male	55	5	325	720
PN01	Male	46	2	128	380
PN03	Male	54	2	244	700
PN05	Female	51	3	104	261
PN06	Male	36	5	282	836
PN07	Female	20	1	62	154
PN09	Female	27	3	203	801
PN10	Male	25	10	365	639
PN11	Female	58	1	55	183
PN12	Male	71	4	290	653
PN13	Female	34	3	264	963
PN14	Male	49	4	163	404
PN16	Female	41	2	230	463
PN17	Male	42	2	153	510
Total	—	—	47	2868	7667

ensure the accuracy of subsequent analyses. First, a bandpass filter was applied to the data with a spectral range of 0.5–45 Hz [34], removing noise and artifacts outside this range, which is typical for EEG signals. Independent component analysis [35] was then applied to remove ocular and muscular artifacts, which are common sources of noise in EEG recordings. Finally, the processed EEG signals were normalized and standardized to remove amplitude variations across different channels and patients [36], ensuring consistency. These preprocessing steps help preserve the integrity of the EEG signals and improve the accuracy and robustness of the seizure detection model.

3.3. Implementation details

As part of this section, a 10-fold cross-validation strategy was adopted to assess the proposed seizure detection method. In this approach, the dataset was divided into 10 folds, with 9 folds used as the training set and 1 fold as the test set. From the training set, 10% was randomly allocated as the validation set to guide model training and prevent overfitting. The optimal hyperparameters were determined using a grid search approach, with specific settings detailed in table 3. To mitigate overfitting, an early stopping strategy was adopted, terminating training when the validation loss shows no further decrease, with a maximum of 50 epochs and a patience value of 5. Based on the highest *F1* score on the validation set, the optimal model was selected, ensuring the best balance between precision and sensitivity. The entire data for 10-fold cross-validation was sourced from the following extraction process: all seizure data were comprehensively extracted based on recorded seizure time segments, and non-seizure data were obtained by randomly selecting a duration 2–4 times [37]

Table 3. Hyperparameters for grid search.

Parameter	Search range	CHB-MIT	Siena
Learning rate	{0.0005, 0.005, 0.001, 0.01}	0.005	0.001
Batch size	{8, 16, 32, 64}	32	32
Attention heads	{2, 4, 8, 16}	4	4
Hidden dim	{32, 64, 128, 256}	64	64
Dropout rate	{0.3, 0.5, 0.7}	0.5	0.5

Table 4. Subject-independent seizure detection SzCORE results on the CHB-MIT dataset.

Dataset	Metric	Event level	Sample level
CHB-MIT	<i>F1</i> -score (%)	66.3	60.8
	Sensitivity (%)	72.7	49.1
	Precision (%)	68.9	71.3
	FP/24 h	41.5	31.6
Siena	<i>F1</i> -score (%)	63.5	28.9
	Sensitivity (%)	58.1	39.2
	Precision (%)	59.8	70.8
	FP/24 h	36.3	24.0

the total seizure duration from corresponding non-seizure segments. Details of the data are provided in tables 1 and 2. Additionally, we validated our approach using the SzCORE framework [38], a standardized open-source research evaluation framework designed to ensure consistent validation of EEG-based seizure detection algorithms across datasets, formats, and performance metrics. Our approach employed both event-based and sample-based evaluations to comprehensively assess performance in subject-independent seizure detection. Specifically, it surpassed the state-of-the-art subject-independent XGBoost model reported in [38], which achieved *F1*-scores of 66.0% for event-based and 52.0% for sample-based evaluations on the CHB-MIT dataset. Detailed results are provided in table 4.

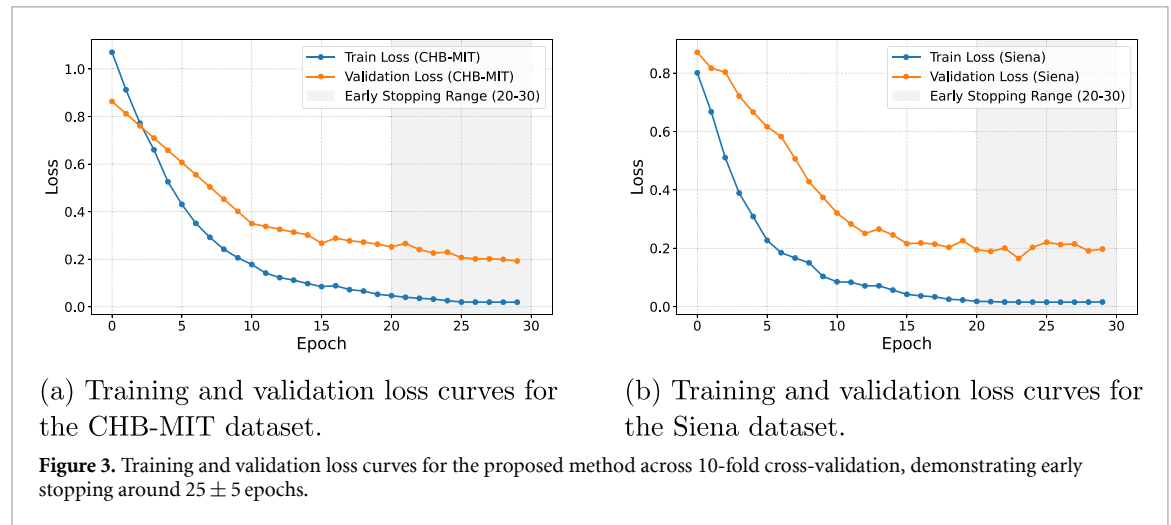
4. Results

4.1. Overall results

In this section, overall results are presented in table 5. The proposed method achieves an accuracy of 98.82% and 98.89% on the CHB-MIT and Siena datasets, respectively. Additionally, it demonstrates high performance across other metrics, with an *F1* score of 98.15% and 98.46%, sensitivity of 98.96% and 98.98%, precision of 98.88% and 98.87%, and a false positive rate per hour (FPR/h) [39] of 0.108 and 0.099 on the CHB-MIT and Siena datasets, respectively. The training and validation loss curves, shown in figure 3, provide robust evidence of the model's stability and convergence across both the CHB-MIT and Siena datasets. These curves illustrate a smooth and consistent reduction in loss values throughout the training process, highlighting the model's ability

Table 5. Overall performance for 10-fold cross-validation.

Dataset	Accuracy (%)	F1 Score (%)	Sensitivity (%)	Precision (%)	FPR/h
CHB-MIT	98.82	98.15	98.96	98.88	0.108
Siena	98.89	98.46	98.98	98.87	0.099



to effectively learn from the data. The early stopping mechanism, which is typically triggered around 25 ± 5 epochs, further underscores the efficiency of the training procedure by preventing overfitting and ensuring optimal generalization. This consistent convergence behavior across the 10-fold cross-validation suggests that the proposed method is well-tuned to handle the variability inherent in the datasets. The gradual decline in loss, particularly noticeable after the initial epochs, reflects the model's capacity to capture underlying patterns, while the plateauing effect near the stopping range highlights a balanced trade-off between model complexity and predictive performance.

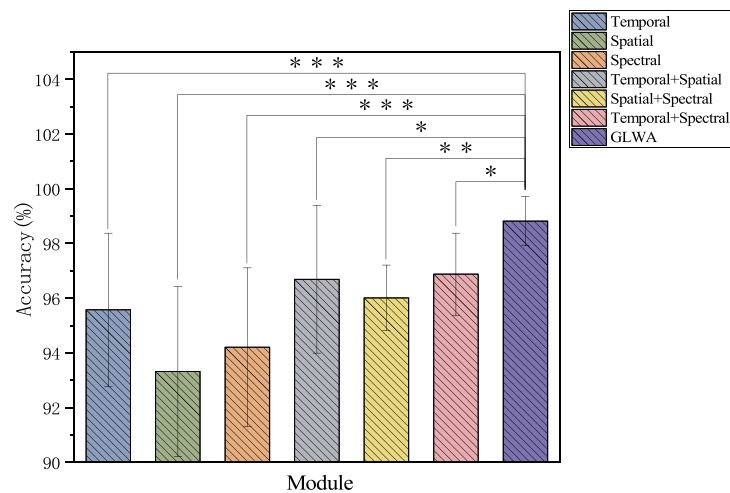
4.2. Ablation experiment

In the ablation experiments performed on the CHB-MIT and Siena datasets, the contribution of each feature—temporal, spatial, and spectral—to the model performance was highlighted. We compared the performance of individual features with that of GLWA, as shown in figure 4. The accuracy of temporal features in the CHB-MIT dataset was 95.58%, while spatial features achieved slightly lower accuracy, and spectral features performed at 93.32%. In the Siena dataset, temporal features played a key role in seizure detection, with spatial features showing relatively lower accuracy. Spectral features in the Siena dataset achieved an accuracy of about 94.09%, similar to spatial features. We employed the paired t -test to analyze the differences in results from the module ablation experiments. The findings revealed statistically significant variations in accuracy across different module combinations (e.g. temporal, spatial, spectral) on both the CHB-MIT and Siena datasets,

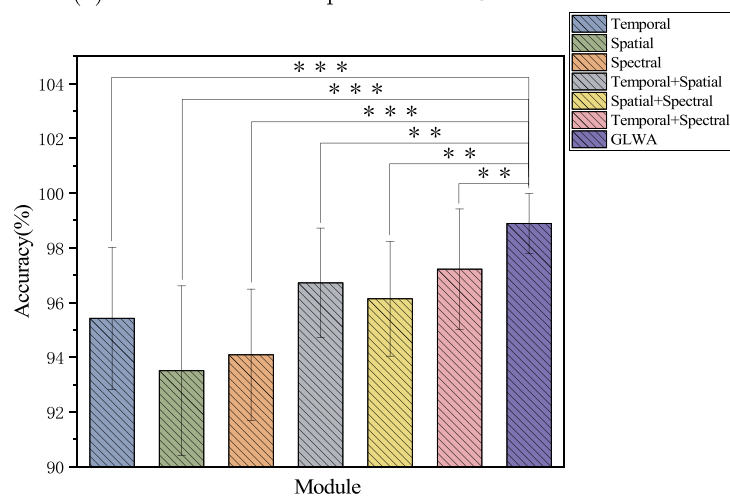
highlighting the substantial contribution of each module to performance enhancement. Moreover, the standard errors for all features remained below 0.04, demonstrating remarkably stable performance across datasets, with the GLWA framework standing out for its exceptional adaptability and precision.

Temporal features contributed the most to seizure detection, followed by spectral and spatial features. All features demonstrated their value across both datasets, with the GLWA model, which integrates temporal, spatial, and spectral features, further enhancing performance. This shows how these features complement each other and strengthen the overall detection capability of the model. Notably, the fusion of temporal and spectral features already brings a significant boost, highlighting the benefits of cross-modal interaction, while the incorporation of spatial information further refines the detection precision.

The performance of GLWA features is excellent on both datasets, allowing for a more comprehensive combination of spatiotemporal and spectral features, thereby improving classification accuracy. GLWA features play a crucial role in the accurate detection of seizures, especially in global and local information fusion. On the CHB-MIT dataset, the accuracy of GLWA model reached 98.82%. On the Siena dataset, it was 98.89%, significantly improving classification accuracy, especially in tasks that require consideration of both spatiotemporal and spectral information. Combining the information of multiple features, especially through the global-local attention mechanism, can significantly improve the performance of the model. In practical applications, proper selection of features and optimization of



(a) Module ablation experiment for CHB-MIT dataset



(b) Module ablation experiment for Siena dataset

Figure 4. The ablation experiments evaluate the contribution of different modules across various datasets. Statistical significance is indicated as follows: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

models are very important to improve classification accuracy.

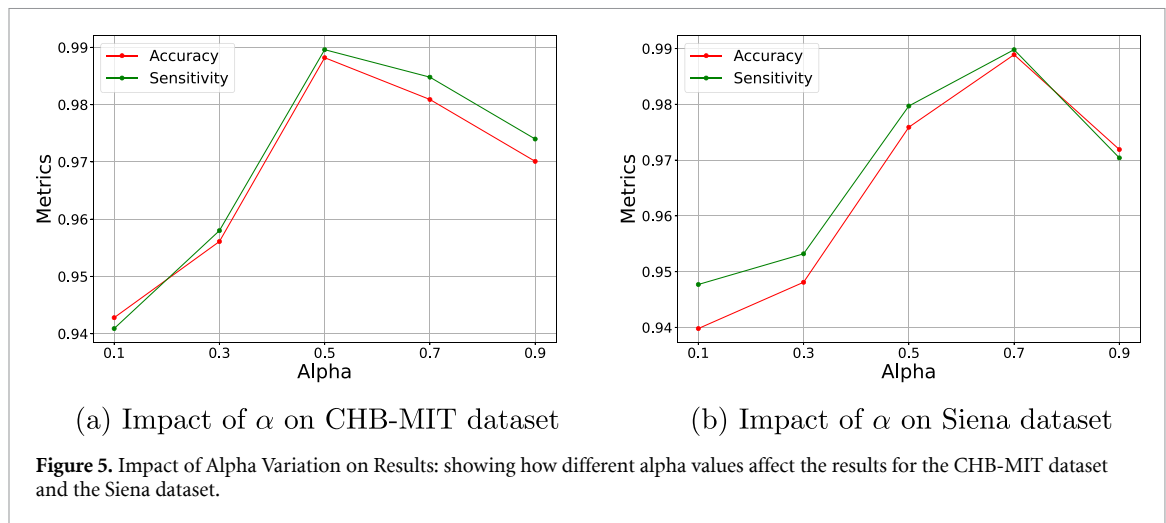
4.3. The influence of parameter α

In this experiment, we controlled the parameter α , which determines the balance between local and global attention, as shown in figure 5. The adjustment of α has a significant impact on the model's performance. Choosing the right value of α can significantly improve classification results, especially when balancing global and local features. When α was too low, the model relied too heavily on local information, whereas when α was too high, the model was prone to interference from global information, affecting classification accuracy. In conclusion, selecting an appropriate α value is crucial for improving the model's classification performance, and the optimal α value differs across datasets.

By adjusting the value of α , we observed the effects of different balances between local and global

features on the model's performance. When α was set between 0.1 and 0.5, there was a significant improvement in both accuracy and sensitivity, showing a strong positive correlation. Yet, as α approached 0.9, both accuracy and sensitivity started to decline, indicating that excessively high values of α negatively impacted performance. Through multiple trials, we found that setting α between 0.5 and 0.7 yielded the best overall results. In this range, the balance between local and global attention effectively captured the key features of seizures, leading to the best classification outcomes. Specifically, α values around 0.7 performed particularly well on the CHB-MIT dataset, while a moderate value of α (around 0.5) provided more stable performance on the Siena dataset.

Despite the similar trends across both datasets, the Siena dataset showed a more significant response to global attention, indicating that the signal structure in the Siena dataset may be more suited for integrating global information.

**Table 6.** Performance Metrics by sliding window of local attention.

Sliding window Size	Accuracy (%)	F1 Score (%)	Sensitivity (%)	Precision (%)
CHB-MIT				
1×1	97.56	96.28	96.79	96.05
2×2	98.44	97.61	97.99	97.51
3×3	98.82	98.15	98.96	98.88
4×4	96.00	96.39	96.10	96.67
5×5	90.18	90.72	90.98	92.32
Siena				
1×1	96.62	95.92	96.77	96.95
2×2	98.89	98.46	98.98	98.87
3×3	98.01	97.00	97.04	97.96
4×4	94.53	94.96	94.94	95.08
5×5	88.88	89.98	90.11	90.05

4.4. The influence of sliding window size for local attention

In this experiment, we examined the model's sensitivity to the sliding window size for local attention by selecting window sizes from 1×1 , 2×2 , 3×3 , 4×4 , 5×5 and evaluating the corresponding classification performance. A window that is too small may fail to capture enough local features, while a window that is too large may lose detailed information. According to the results in table 6, the sliding window sizes of 3 and 2 achieved the best classification performance on the CHB-MIT and Siena datasets, respectively. The optimal sliding window sizes are indicated in bold. These window sizes strike the optimal balance, preserving both local and global features. They are capable of capturing sufficient key signals within shorter time steps while avoiding the interference of excessive background information, thus enhancing the model's focus on critical features for local attention. Additionally, this finding offers valuable insights for model tuning in practical applications, ensuring better classification results across different datasets by

optimizing the window size based on the specific data features.

Although smaller windows typically capture local signal variations, they do not necessarily perform the best in all cases. For instance, in the Siena dataset, the accuracy of the 1×1 window is 96.62%, which is relatively high but still lower than the accuracy of the 3×3 window, which achieved 98.82%. More importantly, the 1×1 window performs slightly worse in terms of sensitivity and F1 score. When the window size becomes too large, the model's performance gradually declines, and it may fail to accurately capture the rapid changes in seizure activity. The increase in window size may also introduce background noise or irrelevant information, especially in parts of the EEG signal with significant background fluctuations. A window that is too large dilutes the importance of the signal, leading to incorrect classifications. For datasets with higher noise or more gradual seizures, using smaller windows can improve sensitivity to signal details. In cases with more frequent seizures and rapid signal changes, a 3×3 window can better

balance information capture and noise suppression for local attention.

5. Discussion

5.1. Analysis of specific seizure segments

5.1.1. Heatmap highlighted by different features and GLWA.

As shown in figure 7(a), in order to further understand the decision-making process of the model in seizure detection, we conducted a detailed analysis of 1 s seizure data of a specific patient in the CHB-MIT dataset. The 1 s data consisted of 256 time steps, with no signs of a seizure in the first 128 time steps, followed by the next 128 time steps where the seizure began at the 128th time step.

We performed a visualization analysis of the temporal, spatial, and spectral modalities, along with the GLWA model. As shown in figure 6, we visualized the regions focused on during a seizure by both different features and GLWA. It can be observed that GLWA integrates the information from individual features, resulting in a more comprehensive focus on the regions during a seizure.

Specifically, the temporal modality primarily concentrated on the signal changes around the temporal point when the seizure occurred, the spatial modality highlighted the relationships between different EEG channels, and the spectral modality revealed the activities in various spectral bands. However, there are certain limitations when relying on each modality individually. The temporal modality can reflect the rapid changes in EEG signals but may overlook longer temporal information in the spatial and spectral domains. The spatial modality cannot capture the instantaneous dynamics of the signal, potentially missing critical changes during seizures. The spectral modality, while effective at revealing frequency activity, lacks detailed information in the time domain and cannot clearly pinpoint the exact moments when seizures start or end.

Additionally, we visualized how the GLWA model integrates these features, showing how different modalities influence the model's focus on specific regions. Through this visualization, we can intuitively observe how the model identifies critical brain regions and features essential for seizure detection, demonstrating the distinct focus of each feature and how the fusion of spatiotemporal and spectral information enhances the model's attention on key areas. The GLWA fusion modality displays the weighted fusion results of temporal, spatial, and spectral information. This combination not only retains the advantages of each modality but also optimizes signal selection and weighting through global and local attention mechanisms, thereby improving seizure detection

accuracy. Compared to using temporal, spatial, or spectral information individually, the GLWA model is able to more comprehensively capture different features of seizures, enhancing its robustness and accuracy.

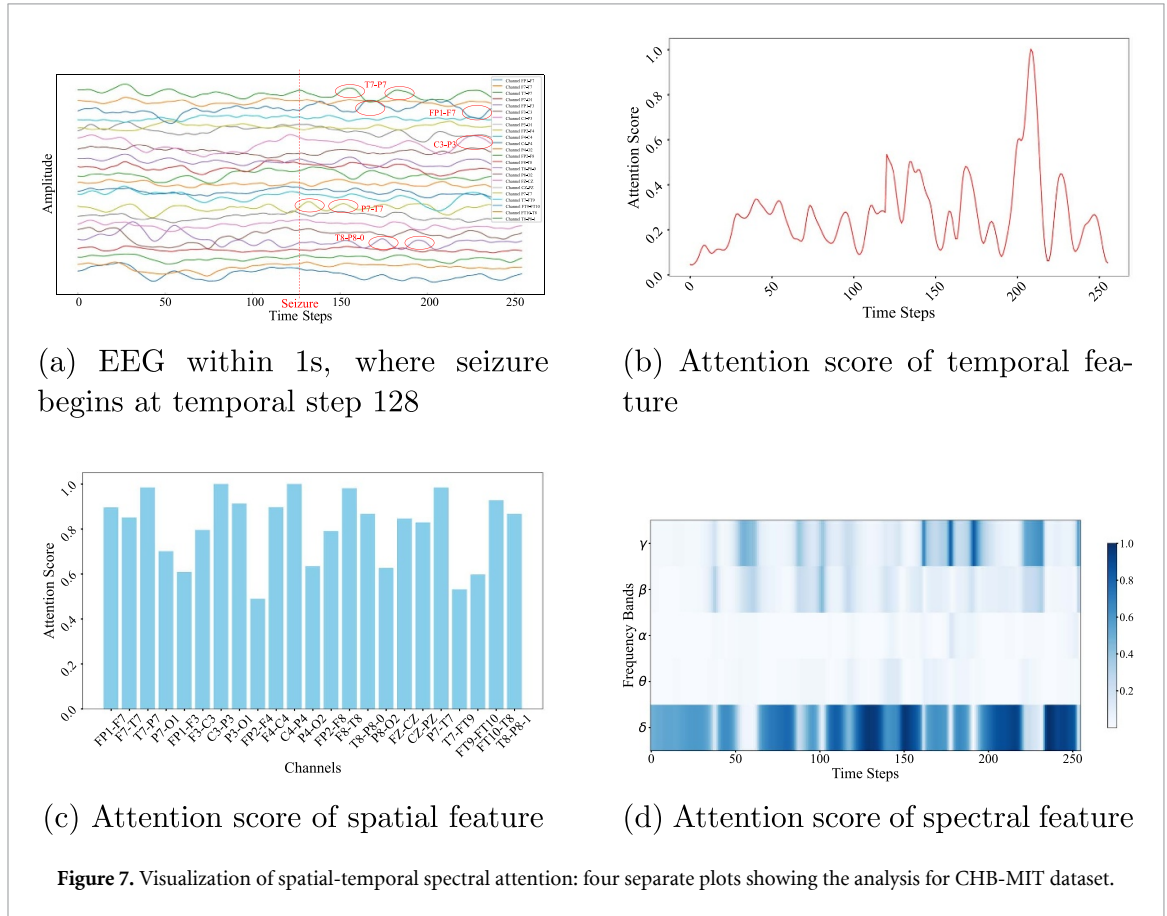
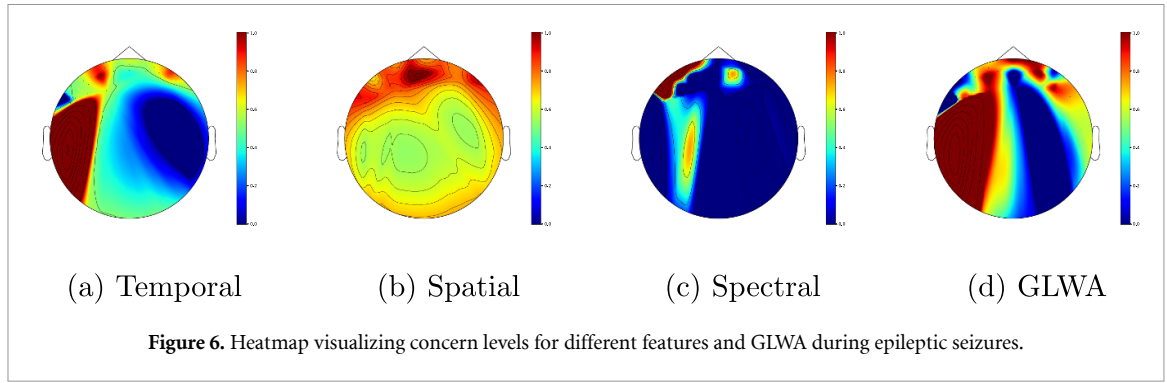
5.1.2. Attention analysis of individual features and GLWA

Meanwhile, as shown in figure 7, by analyzing the temporal and spectral features of the data, we were able to more clearly observe the changes in EEG signals before and after the seizure. The variation in attention scores (figure 7(b)) across different time steps indicated that the scores during the last 128 time steps were significantly higher than those of the first 128 time steps, with the highest attention score occurring around the 210th time step, marking the peak of the seizure. The EEG signals reveal that prior to the seizure, the EEG waveform remains relatively stable. However, once the seizure begins, there are sudden fluctuations in the signals from specific brain regions, particularly at the 210th time step, where the signal rises sharply, indicating obvious seizure activity.

Spatial analysis showed that the C3-P3, T8-P8-0, T7-P7, and FP1-F7 channels had higher attention weights, suggesting that these channels played a significant role in the model's decision-making during the seizure. The amplitude of the brain waves in figure 7(c) further confirms that these channels exhibit significant fluctuations after the 128th time step, highlighting their importance in seizure detection. These channels may be closely related to the transmission pathways of seizures, especially when local brain activity rapidly spreads to other regions at the onset of a seizure. By capturing these fluctuations, the model is able to accurately identify the critical stages of the seizure.

The spectral analysis (figure 7(d)) reveals that δ waves (0.5–4 Hz) contribute more significantly during seizures, especially in their early stages, playing a key role in helping the model accurately detect the onset of the seizure. The model's attention mechanism consistently assigns higher scores to the δ band due to its pronounced power increases and distinct patterns in EEG signals during seizures, which provide strong discriminative features for seizure detection. δ wave activity is often associated with synchronized activity in deep brain structures, which is typical during seizures [40], particularly at their onset. By focusing on these spectral changes, the model can respond more promptly to signal changes, thereby enhancing the sensitivity of seizure detection.

All attention scores have been normalized to ensure consistent comparisons across different time steps. This normalization ensures that we can fairly compare the focus of the model at different time steps, thus verifying its accuracy in seizure recognition.



5.2. Model generalization and sampling strategy optimization

5.2.1. Generalization via leave-one-patient-out cross-validation (LOPOCV)

To comprehensively evaluate the model's generalization ability on unseen subjects [41], we employed LOPOCV [42], which simulates real-world scenarios where the model must adapt to new individuals. This approach ensures strict patient independence, addressing potential data leakage issues inherent in less rigorous validation strategies. By isolating each patient's data as a test set, LOPOCV effectively eliminates cross-contamination between training and testing phases. For a dataset containing n patients, each iteration trains the model on the EEG data of $n - 1$ patients, with the entire EEG data of the remaining patient reserved as the test set. From the training

set, 10% is randomly allocated as the validation set to guide model training and prevent overfitting, while maintaining patient independence. The data are sourced from the extraction process detailed in tables 1 and 2, with learning rate and other parameters kept consistent with prior analyses. Unlike the previously discussed 10-fold cross-validation, LOPOCV employs the complete data of each patient as a test set, offering a more reliable evaluation of the model's ability to generalize to new patients in real-world clinical environments. Detailed test data durations for each patient are provided in tables 7 and 8, and the combined duration of all test sets equals the total recorded duration of each dataset.

The LOPOCV results in tables 7 and 8 demonstrate robust performance for the CHB-MIT and Siena datasets, respectively. For the CHB-MIT

Table 7. Performance metrics of CHB-MIT dataset on testing data with LOPOCV.

Case	Testing data (h)	Accuracy (%)	F1 Score (%)	Sensitivity (%)	Precision (%)	FPR/h
chb01	40.55	89.23	85.12	87.89	88.34	0.130
chb02	35.27	87.65	83.98	89.03	86.78	0.125
chb03	38.00	94.19	93.48	95.02	92.97	0.122
chb04	156.07	77.89	77.92	82.50	81.67	0.133
chb05	39.00	87.34	83.21	88.78	86.45	0.131
chb06	66.74	89.78	85.67	87.12	88.56	0.124
chb07	67.05	94.51	92.68	93.82	91.43	0.128
chb08	20.00	93.44	92.87	93.74	92.78	0.120
chb09	67.87	87.89	83.45	87.67	86.90	0.133
chb10	50.02	89.01	85.23	88.90	87.45	0.126
chb11	34.79	88.34	84.78	87.23	88.67	0.129
chb12	20.69	90.23	86.12	89.45	87.89	0.121
chb13	33.00	94.14	93.75	92.52	94.33	0.132
chb14	26.00	89.45	85.56	87.89	88.23	0.125
chb15	39.00	88.78	84.45	88.67	87.12	0.130
chb16	19.00	90.89	86.78	87.34	89.45	0.119
chb17	21.01	95.72	94.33	95.58	94.16	0.134
chb18	34.63	89.67	85.90	88.23	88.90	0.123
chb19	28.93	88.23	84.12	87.56	87.34	0.128
chb20	27.60	92.49	92.95	94.94	93.22	0.120
chb21	32.83	87.90	83.67	88.45	86.89	0.132
chb22	31.00	89.12	85.23	87.78	88.01	0.126
chb23	26.56	88.56	84.89	89.34	87.67	0.124
chb24	21.30	90.01	86.01	88.90	89.23	0.121
Average	40.70	89.84	86.76	89.52	88.94	0.127

Table 8. Performance metrics of Siena dataset on testing data with LOPOCV.

Case	Testing data (min)	Accuracy (%)	F1 Score (%)	Sensitivity (%)	Precision (%)	FPR/h
PN00	198	90.34	83.45	87.12	88.90	0.125
PN01	809	91.23	82.78	88.56	89.34	0.115
PN03	752	87.89	84.23	86.78	87.45	0.122
PN05	359	88.56	86.12	85.90	88.23	0.118
PN06	722	89.78	83.90	88.90	86.78	0.124
PN07	523	90.12	85.67	87.34	89.01	0.116
PN09	410	79.18	84.90	86.45	84.41	0.112
PN10	1002	92.34	90.41	92.76	88.97	0.113
PN11	145	87.67	86.34	85.67	87.12	0.127
PN12	246	89.01	83.12	88.23	86.90	0.121
PN13	519	90.89	85.01	87.78	89.45	0.117
PN14	1408	91.56	82.89	89.34	88.01	0.114
PN16	303	88.23	86.78	86.01	87.56	0.126
PN17	308	89.45	84.56	88.90	87.23	0.119
Average	550	89.75	84.68	87.56	87.98	0.119

dataset, the average accuracy is 89.94%, $F1$ score is 86.76%, sensitivity is 89.52%, precision is 88.94%, and false positive rate per hour (FPR/h) is 0.127. For the Siena dataset, the average accuracy reaches 89.75%, $F1$ score is 84.68%, sensitivity is 87.56%, precision is 87.98%, and FPR/h is 0.119. These average results are highly commendable, reflecting the model's strong capability to generalize effectively to unseen patients despite the inherent challenges posed by inter-patient variability in EEG signal features. Notably, in the CHB-MIT dataset, all cases except chb04 (with an accuracy of 77.89%) achieve an accuracy of 80% or higher, while in the Siena

dataset, all cases except PN09 (with an accuracy of 79.18%) also exceed 80%, highlighting the consistency of the model's performance across most patients. As demonstrated in table 9, our GLWA method exhibits superior performance compared to other patient-independent approaches on the CHB-MIT dataset, which is indicated in bold. In comparison with Aboyeji *et al* [42], which reported an accuracy of 87.29% and a sensitivity of 85.36%, our approach achieves notable enhancements in both accuracy and sensitivity. The robustness and adaptability of our GLWA method further solidify its effectiveness in diverse clinical scenarios. This improvement indicates

Table 9. Comparison of performance metrics across different studies on patient-independent evaluations on the CHB-MIT dataset.

Study	Year	Method	Accuracy (%)	Sensitivity (%)
Zhang <i>et al</i> [43]	2020	LOPOCV	85.44	88.04
Yao <i>et al</i> [44]	2021	LOPOCV	83.89	83.72
Aboyaji <i>et al</i> [42]	2025	LOPOCV	87.29	85.36
Mohamed <i>et al</i> [45]	2025	LOPOCV	—	80.70
GLWA	2023	LOPOCV	89.84	89.52

Table 10. Performance metrics for different non-seizure sampling strategies.

Dataset	Strategy	Accuracy (%)	F1 Score (%)	Sensitivity (%)	Precision (%)	FPR/h
CHB-MIT	Random	89.94	86.76	89.52	88.94	0.127
	Far	90.97	88.05	89.00	93.42	0.055
	Near	87.03	84.19	85.64	83.73	0.148
Siena	Random	89.02	85.01	87.84	87.81	0.119
	Far	91.79	87.61	87.85	94.10	0.083
	Near	85.81	81.99	84.08	84.09	0.126

that our approach better captures patient-specific EEG patterns, enhancing generalization in realistic clinical scenarios. These findings underscore the importance of LOPOCV in evaluating model robustness and suggest that further improvements, such as patient-specific feature adaptation or advanced domain generalization techniques, could enhance performance in practical applications.

5.2.2. Model generalization and sampling strategy optimization

The LOPOCV results presented in tables 7 and 8 employ a random sampling strategy, selecting non-seizure samples randomly from files without seizure segments. This approach enhances sample diversity by incorporating varied temporal intervals, thereby improving the model's generalization across different patients' EEG signals. West *et al* [46] noted that temporal separation between seizure and non-seizure files may influence prediction accuracy. To further explore this aspect, we evaluated three non-seizure sampling strategies:

- Random sampling: non-seizure samples are randomly selected from files without seizure segments, regardless of temporal distance from seizure events.
- Far sampling: non-seizure samples are chosen to be temporally non-continuous with seizure events.
- Near sampling: non-seizure samples are selected to be temporally continuous with seizure events.

Table 10 presents the performance metrics for these strategies across the CHB-MIT and Siena datasets. Random sampling achieves stable performance, with an accuracy of 89.94% on CHB-MIT and 89.02% on Siena, reflecting its ability to capture diverse EEG signal patterns through varied temporal distributions. This strategy supports robust

model training suitable for broad clinical applications. Far sampling improves both accuracy and precision, reaching 90.97% accuracy and 93.42% precision on CHB-MIT, and 91.79% accuracy on Siena. This approach enhances the model's ability to distinguish between seizure and non-seizure states, particularly in tasks requiring high discriminative power. Near sampling yields slightly lower accuracies of 87.03% on CHB-MIT and 85.81% on Siena, focusing on EEG signal differences between seizure and non-seizure periods. By prioritizing samples close to seizure events, this strategy aligns with real-world clinical scenarios, enhancing model robustness for practical applications.

Our method demonstrates strong performance across all three sampling strategies, showcasing its versatility and effectiveness in EEG signal analysis. The consistent results across random, far, and near sampling highlight the model's ability to adapt to varying temporal constraints while maintaining high accuracy and reliability. This adaptability ensures effective handling of diverse clinical requirements, from broad generalization to precise differentiation and clinically relevant scenarios, establishing our approach as a robust solution for seizure prediction tasks.

5.3. Comparison with other advanced methods

We compare the performance of our proposed GLWA method with existing methods on CHB-MIT datasets (table 11) and Siena datasets (table 12). The results show that the proposed GLWA method is superior to other methods in terms of accuracy and sensitivity, which is indicated in bold. This indicates that GLWA method has strong potential and effectiveness in the diagnosis of epilepsy. Compared with Chen *et al* [47] -spiking conformer, the accuracy and sensitivity of the GLWA model on the CHB-MIT dataset improve

Table 11. Comparison with other methods on CHB-MIT dataset.

Authors	Year	Method	Validation method	Accuracy (%)	Sensitivity (%)
Li <i>et al</i> [14]	2020	CNN+LSTM	—	95.29	95.42
Zhang <i>et al</i> [43]	2020	AttVGGNet-RC	8-fold CV	95.12	94.62
Hu <i>et al</i> [49]	2020	LMD+Bi-LSTM	—	—	93.61
Yao <i>et al</i> [44]	2021	Attention+Bi-LSTM	10-fold CV	87.80	87.30
Alturki <i>et al</i> [50]	2021	CSP-LBP+KNN	5-fold CV	98.62	—
Shen <i>et al</i> [51]	2022	DWT+RUSBoosted tree	5-fold CV	96.38	96.15
He <i>et al</i> [52]	2022	GAT+Bi-LSTM	5-fold CV	98.52	97.75
Zhang <i>et al</i> [37]	2022	Bi-GRU	10-fold CV	98.49	93.89
He <i>et al</i> [52]	2022	GAT+BiLSTM	5-fold CV	98.52	97.75
Xiong <i>et al</i> [53]	2022	Improved Genetic Algorithm	5-fold CV	97.26	96.89
Zhang <i>et al</i> [54]	2024	HGO+SVM	5-fold CV	98.63	98.06
Chen <i>et al</i> [47]	2024	Spiking Conformer	10-Fold CV	97.10	94.90
Fu <i>et al</i> [48]	2025	WCE-MASE-PGCN	LOOCV	—	97.90
GLWA	2025	GLWA	10-fold CV	98.82	98.96

Table 12. Comparison with other methods on Siena dataset.

Author(s)	Year	Model	Validation method	Accuracy (%)	Sensitivity (%)
Xiong <i>et al</i> [53]	2022	Improved Genetic Algorithm	5-fold CV	98.88	98.36
Zhao <i>et al</i> [22]	2023	CNN+Transformer	10-fold CV	97.87	97.40
Dokare <i>et al</i> [55]	2025	SA+RF	5-fold CV	94.81	94.04
GLWA	2025	GLWA	10-fold CV	98.89	98.98

by 1.72% and 4.04%, respectively. Compared with the leave-one-out cross-validation (LOOCV) method proposed by Fu *et al* [48], our proposed GLWA model demonstrates better performance, which further confirms that our model outperforms other methods across various scenarios. On the Siena dataset, compared with Zhao *et al* [22], the accuracy and sensitivity of GLWA model increased by 1.02% and 1.58%. The GLWA model demonstrated outstanding performance on both datasets, significantly outperforming other methods in terms of accuracy and sensitivity, highlighting its effectiveness on complex datasets. Combining the global-local attention mechanism can greatly improve the classification performance of the model, and its comprehensive performance is the most outstanding.

Meanwhile, compared with table 4, the lower performance in SzCORE's table 4 reflects its stricter evaluation framework, which ensures robust subject-independent assessment by maintaining complete separation of training and test sets across patients, resulting in lower scores. In contrast, tables 11 and 12 utilize 10-fold cross-validation, a standard approach known for producing stable and generalizable performance estimates, which can be viewed as the upper performance limit achievable with minimal calibration for clinical application. SzCORE's rigorous approach guarantees reliable generalization across diverse patient profiles, essential for real-world clinical reliability, while the 10-fold method in tables 11 and 12 emphasizes precision in similar data distributions, offering complementary insights for clinical translation.

6. Conclusion

The GLWA model for seizure detection integrates spatial, temporal, and spectral feature fusion, significantly enhancing detection accuracy. By effectively combining these features, the model captures a comprehensive view of EEG signals, improving its ability to identify seizures. Through comprehensive experiments on the CHB-MIT and Siena datasets, the GLWA model demonstrates superior performance across multiple evaluation metrics, surpassing existing methods in terms of sensitivity and overall accuracy. These results highlight the model's exceptional performance in seizure detection, showing that it can detect seizures with high precision even in complex, noisy data. Future research could further enhance the model's robustness and generalization capabilities by incorporating multimodal learning and adaptive fusion strategies. Addressing challenges posed by heterogeneous data, such as variations across different patients and clinical conditions, and improving clinical applicability would also be key to ensuring the model's success in real-world settings. By refining the model to adapt to a broader range of data and environments, it could become a more reliable tool for seizure detection in clinical practice.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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